Equations with Brackets

AQA · KS3 Year 8

Name: ____________________ Class: __________ Date: __________

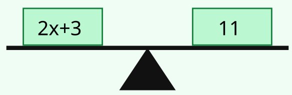
Solve linear equations that contain brackets. Expand the brackets first by multiplying every term inside, then collect like terms and solve as usual.

VISUAL EXAMPLE

$2x + 3 = 11$

-3 both sides $2x = 8$

÷2 both sides $x = 4$



An equation balances: do the same to both sides to keep it level.

COMMON MISTAKES

- Only multiplying the first term inside the bracket: $3(x + 2) = 3x + 6$, not $3x + 2$.
- Forgetting that a negative outside the bracket changes both signs inside.
- Trying to divide by the number outside before expanding when other terms are present.
- Sign errors when subtracting an expanded bracket such as $-2(x - 4)$.

METHOD

1. Expand each bracket — multiply the term outside by every term inside.
2. Collect like terms on each side.
3. If letters appear on both sides, move the smaller letter term across.
4. Collect the numbers, then divide to find the letter.
5. Check by substituting into the original (un-expanded) equation.

WORKED EXAMPLE

Solve: $3(x + 2) = 18$

Step 1 — Expand: $3 \times x + 3 \times 2 = 3x + 6$, so $3x + 6 = 18$.

Step 2 — (Already collected.)

Step 3 — (No letters on the right.)

Step 4 — Subtract 6: $3x = 12$; divide by 3: $x = 4$.

Step 5 — Check: $3(4 + 2) = 3 \times 6 = 18$

1 Solve. (2 marks)

a) $2(x + 3) = 14$
b) $5(x - 1) = 20$

HOW TO START

Expand the bracket first: multiply the outside number by each term inside.
For $2(x + 3) = 14$, this gives $2x + 6 = 14$, then $2x = 8$, $x = 4$.

ANSWERS

(a) = ______________________

(b) = ______________________

METHOD 1 Expand every bracket fully | 2 Collect like terms on each side | 3 Move the smaller letter term across | 4 Collect numbers, then divide | 5 Check in the original equation

2 Solve. (2 marks)

a) $3(x + 4) = 18$

b) $4(2x - 1) = 28$

HOW TO START

Expand, then undo the + or - , then the ×.

For $4(2x - 1) = 28$: $8x - 4 = 28$, so $8x = 32$, $x = 4$.

ANSWERS

(a) = ____________________

(b) = ____________________

METHOD 1 Expand every bracket fully | 2 Collect like terms on each side | 3 Move the smaller letter term across | 4 Collect numbers, then divide | 5 Check in the original equation

3 Solve. (3 marks)

a) $2(3x + 1) = 26$

b) $5(2x - 3) = 35$

c) $3(x + 5) = 12$

ANSWERS

(a) = ____________________

(b) = ____________________

(c) = ____________________

METHOD 1 Expand every bracket fully | 2 Collect like terms on each side | 3 Move the smaller letter term across | 4 Collect numbers, then divide | 5 Check in the original equation

4 Solve. Letters appear on both sides. (3 marks)

a) $2(x + 4) = x + 11$

b) $4(x - 1) = 2(x + 3)$

c) $3(2x + 1) = 4x + 11$

ANSWERS

(a) = ____________________

(b) = ____________________

(c) = ____________________

METHOD 1 Expand every bracket fully | 2 Collect like terms on each side | 3 Move the smaller letter term across | 4 Collect numbers, then divide | 5 Check in the original equation

5 Expand, collect like terms, then solve. (3 marks)

a) $2(x + 1) + 3(x - 2) = 11$

b) $5(x - 1) - 2(x - 4) = 18$

c) $4(x + 2) - 3(x - 1) = 15$

ANSWERS

(a) = __________

(b) = __________

(c) = __________

METHOD 1 Expand every bracket fully | 2 Collect like terms on each side | 3 Move the smaller letter term across | 4 Collect numbers, then divide | 5 Check in the original equation

6 A rectangle has length (x + 5) cm and width 3 cm. (3 marks)
The perimeter is 28 cm.

a) Form an equation using 2(length + width). (1 mark)
b) Solve it to find x. (1 mark)
c) Work out the area of the rectangle. (1 mark)

ANSWERS

(a) = ____________________

(b) = ____________________

(c) = ____________________

METHOD 1 Expand every bracket fully | 2 Collect like terms on each side | 3 Move the smaller letter term across | 4 Collect numbers, then divide | 5 Check in the original equation

7 A regular pentagon has five equal sides, each $(2x - 1)$ cm long. (4 marks)
The perimeter is 45 cm.

a) Form an equation. (1 mark)
b) Solve it to find x. (2 marks)
c) Find the length of one side. (1 mark)

ANSWERS

(a) = ______

(b) = ______

(c) = ______

METHOD 1 Expand every bracket fully | 2 Collect like terms on each side | 3 Move the smaller letter term across | 4 Collect numbers, then divide | 5 Check in the original equation

8 a) Solve $3(2x - 1) = 2(x + 5) + 3$. (2 marks) (5 marks)

b) Show that the equation $4(x + 2) = 2(2x - 1)$ has no solution, and explain what this tells you. (3 marks)

ANSWERS

(a) = ____________________

(b) = ____________________

METHOD 1 Expand every bracket fully | 2 Collect like terms on each side | 3 Move the smaller letter term across | 4 Collect numbers, then divide | 5 Check in the original equation

Self-reflection

How confident do you feel with this topic now? (tick one)

Not yet	Getting there	Confident	Really confident

Which question did you find the hardest, and why?

Why must I multiply every term inside the bracket?

One thing I will practise before the next lesson:

Teacher key

Q1: a) $2x + 6 = 14 \quad 2x = 8 \quad x = 4$ b) $5x - 5 = 20 \quad 5x = 25 \quad x = 5$

Q2: a) $3x + 12 = 18 \quad 3x = 6 \quad x = 2$ b) $8x - 4 = 28 \quad 8x = 32 \quad x = 4$

Q3: a) $6x + 2 = 26 \quad 6x = 24 \quad x = 4$ b) $10x - 15 = 35 \quad 10x = 50 \quad x = 5$ c) $3x + 15 = 12 \quad 3x = -3 \quad x = -1$

Q4: a) $2x + 8 = x + 11 \quad x = 3$ b) $4x - 4 = 2x + 6 \quad 2x = 10 \quad x = 5$ c) $6x + 3 = 4x + 11 \quad 2x = 8 \quad x = 4$

Q5: a) $5x - 4 = 11 \quad 5x = 15 \quad x = 3$ b) $3x + 3 = 18 \quad 3x = 15 \quad x = 5$ c) $x + 11 = 15 \quad x = 4$

Q6: a) $2((x + 5) + 3) = 28$ b) $2(x + 8) = 28 \quad x + 8 = 14 \quad x = 6$ c) length = 11 cm, area = $11 \times 3 = 33$ cm^2

Q7: a) $5(2x - 1) = 45$ b) $10x - 5 = 45 \quad 10x = 50 \quad x = 5$ c) side = $2(5) - 1 = 9$ cm

Q8: a) $6x - 3 = 2x + 13 \quad 4x = 16 \quad x = 4$ b) $4x + 8 = 4x - 2 \quad 8 = -2$, which is impossible, so there is no value of x that works (no solution).